

# WEEK 3: SELLER AND INVENTORY MANAGEMENT SYSTEM

## Scenario

The pet e-commerce company sells pet food, toys, grooming products, accessories, and healthcare products supplied by multiple sellers. The system should maintain seller details, track products supplied by each seller, and monitor inventory levels to ensure that pet products are available for customers.

## 1. Create Seller and Inventory Tables

Seller Table

```
SQL> CREATE TABLE Seller (  
2     seller_id VARCHAR2(10) PRIMARY KEY,  
3     seller_name VARCHAR2(50),  
4     email VARCHAR2(50),  
5     phone VARCHAR2(15)  
6 );  
  
Table created.
```

Inventory Table

```
SQL> CREATE TABLE Inventory (  
2     inventory_id VARCHAR2(10) PRIMARY KEY,  
3     product_id VARCHAR2(10),  
4     stock_quantity NUMBER,  
5     FOREIGN KEY (product_id) REFERENCES Product(product_id)  
6 );  
  
Table created.
```

## 2. Establish Relationships

Seller and Product

One seller can supply multiple pet products.

One pet product can be supplied by one or multiple sellers.

Seller —————< Product  
One to Many

Product and Inventory

Each pet product has inventory information.

Inventory tracks the available stock of each product.

Product ————— Inventory  
One to One

### 3. Primary Key and Foreign Key

#### PRIMARY KEY FOR BOTH THE TABLES:

Seller Table  
seller\_id → Primary Key

Inventory Table  
inventory\_id → Primary Key

```
SQL> ALTER TABLE Seller ADD CONSTRAINT PK_Seller PRIMARY KEY(Seller_ID);  
Table altered.  
  
SQL> ALTER TABLE Inventory ADD CONSTRAINT PK_Inv PRIMARY KEY(Inventory_ID);  
Table altered.
```

#### FOREIGN KEY FOR INVENTORY TABLE:

Inventory Table  
product\_id → Foreign Key  
References Product(product\_id)

```
SQL> ALTER TABLE Inventory ADD  
2 CONSTRAINT FK_Prod FOREIGN KEY(Product_ID) REFERENCES Product(Product_ID);  
Table altered.
```

```
SQL> ALTER TABLE Inventory ADD  
2 CONSTRAINT FK_Seller FOREIGN KEY(Seller_ID) REFERENCES Seller(Seller_ID);  
Table altered.
```

### 4. Constraints

Make Seller Email Unique

```
SQL> ALTER TABLE Seller  
2 ADD CONSTRAINT uq_seller_email UNIQUE(email);  
  
Table altered.
```

Make Email NOT NULL

```
SQL> ALTER TABLE Seller  
2 MODIFY email NOT NULL;  
  
Table altered.
```

Check Stock Quantity is Positive

```
SQL> ALTER TABLE Inventory
  2  ADD CONSTRAINT chk_stock_quantity
  3  CHECK (stock_quantity >= 0);

Table altered.
```

## 5. Insert Values into Seller Table

```
SQL> INSERT INTO Seller VALUES
  2  ('S001', 'PetCare Suppliers', 'petcare@gmail.com', '9876543210');

1 row created.

SQL> INSERT INTO Seller VALUES
  2  ('S002', 'Happy Pets Store', 'happypets@gmail.com', '9876543211');

1 row created.

SQL> INSERT INTO Seller VALUES
  2  ('S003', 'Paws World', 'pawsworld@gmail.com', '9876543212');

1 row created.

SQL> INSERT INTO Seller VALUES
  2  ('S004', 'Pet Zone Distributors', 'petzone@gmail.com', '9876543213');

1 row created.

SQL> INSERT INTO Seller VALUES
  2  ('S005', 'Healthy Paws', 'healthypaws@gmail.com', '9876543214');

1 row created.
```

```
SQL> select * from seller;
```

```
SELLER_ID  SELLER_NAME
```

```
EMAIL
```

```
PHONE
```

```
S001      PetCare Suppliers  
petcare@gmail.com
```

```
9876543210
```

```
S002      Happy Pets Store  
happypets@gmail.com
```

```
9876543211
```

```
S003      Paws World  
pawsworld@gmail.com
```

```
9876543212
```

```
S004      Pet Zone Distributors  
petzone@gmail.com
```

```
9876543213
```

```
S005      Healthy Paws  
healthypaws@gmail.com
```

```
9876543214
```

## 6. Insert Values into Inventory Table

```
SQL> INSERT INTO Inventory VALUES ('I001', 'P101', 25);
1 row created.

SQL> INSERT INTO Inventory VALUES ('I002', 'P102', 30);
1 row created.

SQL> INSERT INTO Inventory VALUES ('I003', 'P103', 40);
1 row created.

SQL> INSERT INTO Inventory VALUES ('I004', 'P104', 35);
1 row created.

SQL> INSERT INTO Inventory VALUES ('I005', 'P105', 25);
1 row created.

SQL> INSERT INTO Inventory VALUES ('I006', 'P106', 15);
1 row created.

SQL> INSERT INTO Inventory VALUES ('I007', 'P107', 15);
1 row created.

SQL> INSERT INTO Inventory VALUES ('I008', 'P108', 18);
1 row created.

SQL> INSERT INTO Inventory VALUES ('I009', 'P109', 12);
1 row created.

SQL> INSERT INTO Inventory VALUES ('I010', 'P110', 10);
1 row created.
```

```
SQL> SELECT * FROM Inventory;
```

| INVENTORY_ | PRODUCT_ID | STOCK_QUANTITY |
|------------|------------|----------------|
| I001       | P101       | 25             |
| I002       | P102       | 30             |
| I003       | P103       | 40             |
| I004       | P104       | 35             |
| I005       | P105       | 25             |
| I006       | P106       | 15             |
| I007       | P107       | 15             |
| I008       | P108       | 18             |
| I009       | P109       | 12             |
| I010       | P110       | 10             |

```
10 rows selected.
```

## 7. Report 1: Seller-wise Product Report

Display: Seller name, Number of products supplied, Available stock

```
SQL> SELECT
  2     c.category_name,
  3     COUNT(p.product_id) AS number_of_products,
  4     SUM(p.stock_quantity) AS available_stock
  5 FROM Category c
  6 JOIN Product p
  7     ON c.category_id = p.category_id
  8 GROUP BY c.category_name;
```

| CATEGORY_NAME   | NUMBER_OF_PRODUCTS | AVAILABLE_STOCK |
|-----------------|--------------------|-----------------|
| pet bedding     | 1                  | 10              |
| fish            | 1                  | 35              |
| pet toys        | 2                  | 40              |
| dogs            | 1                  | 25              |
| pet health      | 1                  | 12              |
| cats            | 1                  | 30              |
| birds           | 1                  | 40              |
| pet grooming    | 1                  | 15              |
| pet accessories | 1                  | 18              |

```
9 rows selected.
```

## 8. Report 2: Stock Availability Report

Display: Product name, Stock quantity, Stock status

```

SQL> SELECT p.product_name,
2         i.stock_quantity,
3         CASE
4             WHEN i.stock_quantity = 0 THEN 'OUT OF STOCK'
5             WHEN i.stock_quantity <= 10 THEN 'LOW STOCK'
6             ELSE 'AVAILABLE'
7         END AS stock_status
8 FROM Product p
9 JOIN Inventory i ON p.product_id = i.product_id;

```

| PRODUCT_NAME           | STOCK_QUANTITY | STOCK_STATUS |
|------------------------|----------------|--------------|
| pedigree dog food      | 25             | AVAILABLE    |
| cat dry food           | 30             | AVAILABLE    |
| bird seed mix          | 40             | AVAILABLE    |
| fish food flakes       | 35             | AVAILABLE    |
| puppy chew toy         | 25             | AVAILABLE    |
| cat ball toy           | 15             | AVAILABLE    |
| pet shampoo            | 15             | AVAILABLE    |
| Dog collar             | 18             | AVAILABLE    |
| pet vitamin supplement | 12             | AVAILABLE    |
| pet sleeping bed       | 10             | LOW STOCK    |

10 rows selected.

### 9. Report 3: Inventory Analysis

Find: Total products available, Products out of stock, Highest stocked products, Average inventory quantity

Total Products Available

```

SQL> SELECT COUNT(*) AS total_products
2 FROM Inventory
3 WHERE stock_quantity > 0;

```

| TOTAL_PRODUCTS |
|----------------|
| 10             |

Products Out of Stock

```

SQL> SELECT COUNT(*) AS out_of_stock_products
2 FROM Inventory
3 WHERE stock_quantity = 0;

```

| OUT_OF_STOCK_PRODUCTS |
|-----------------------|
| 0                     |

## Highest Stocked Products

```
SQL> SELECT p.product_name, i.stock_quantity
  2  FROM Product p
  3  JOIN Inventory i ON p.product_id = i.product_id
  4  WHERE i.stock_quantity = (
  5        SELECT MAX(stock_quantity) FROM Inventory
  6  );
```

| PRODUCT_NAME  | STOCK_QUANTITY |
|---------------|----------------|
| bird seed mix | 40             |

## Average Inventory Quantity

```
SQL> SELECT AVG(stock_quantity) AS average_inventory
  2  FROM Inventory;
```

| AVERAGE_INVENTORY |
|-------------------|
| 22.5              |